

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – TECHNICAL PRESENTATION

Course Code:	ITA03	Course Title:	Mobile computing
CO Assessed:	CO3 – Precision (BL3)	Assessment:	Assessment Tool 2 – Technical Presentation
Weightage:	30 % of CIA	Total Marks:	20 (Answer ANY ONE)
CO3 Statement:	Implement wireless networking and Mobile IP concepts for routing in cellular and ad hoc networks.. (BTL 3)		
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Section / Batch:	D/ 3 rd year	Date:	6.8.26

Code of Conduct

I Rose Mystica (192421410) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- Duration: 10 minutes

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Question

1. Derive the total OFDMA capacity in Mbps for a system where each of 1000 subcarriers supports 4 bits/symbol at 15 kHz subcarrier spacing, using $\text{throughput} \approx \text{subcarriers} \times \text{spacing} \times \text{bits/symbol}$, showing the full calculation. Discuss how this formula reveals three independent levers (subcarrier count, spacing, modulation order) a network can adjust to increase throughput, and the practical limits on each.



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Technical Presentation

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Given:

- Number of subcarriers = **1000**
- Subcarrier spacing = **15 kHz = 15,000 Hz**
- Bits per symbol = **4 bits/symbol**

○ **Formula:**

$$\text{Throughput} = \text{Subcarriers} \times \text{Spacing} \times \text{Bits/Symbol}$$

Calculation:

$$= 1000 \times 15,000 \times 4$$

$$= 60,000,000 \text{ bits/s}$$

$$= \frac{60,000,000}{1,000,000} = 60 \text{ Mbps}$$

Answer:

Total OFDMA Capacity $\approx$ 60 Mbps

Three Independent Levers to Increase Throughput

- Number of subcarriers
- Subcarrier spacing
- Number of bits transmitted per symbol

2. Increase subcarrier spacing

- Currently:

15 kHz

- If we increase it to 30 kHz:

$$1000 \times 30000 \times 4 = 120 \text{ Mbps}$$

But:

Larger spacing requires more bandwidth for the same number of subcarriers and has channel/propagation trade-offs.

3. Increase bits per symbol

Currently:

4 bits/symbol

- If we use 6 bits/symbol:

$$1000 \times 15000 \times 6 = 90 \text{ Mbps}$$

But:

Higher bits/symbol means higher-order modulation, which requires **better signal quality**